# Supplemental Material for Papers 3 and 4:
# Dark Matter as Information Loss and the Unified $\Sigma$ Framework

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# Part I: Paper 3 — Dark Matter from the Gravitational Channel

## I. S1. RUNNING $G$ DERIVATION FROM THE GRAVITATIONAL CHANNEL

### A. The scale-dependent gravitational channel

The gravitational channel at momentum scale $k$ is defined as

$$\mathcal{N}_{\text{grav}}(k) : \rho \mapsto \text{Tr}_{E(k)}\big[U(G(k))\,\rho\,U(G(k))^\dagger\big], \tag{1}$$

where $U(G(k))$ is the unitary evolution with the Hamiltonian $H(G(k))$ and the environment trace represents gravitational decoherence at scale $k$. The entropy production at scale $k$ is

$$\Sigma(k) = D(\rho\|\sigma) - D\big(\mathcal{N}_{\text{grav}}(k)(\rho)\,\big\|\,\mathcal{N}_{\text{grav}}(k)(\sigma)\big). \tag{2}$$

By the data processing inequality (DPI), $\Sigma(k) \geq 0$ for all $k$. For scales $k_{\text{IR}} < k_{\text{UV}}$, coarse-graining from UV to IR constitutes an additional channel, giving the monotonicity

$$\Sigma(k_{\text{IR}}) \geq \Sigma(k_{\text{UV}}) \geq 0. \tag{3}$$

This is the Casini–Huerta result [1]: the renormalization group (RG) flow is a quantum channel, and the QRE decreases monotonically under the RG flow.

### B. Kumar's QFT first-principles running

Following Kumar [2], the gravitational coupling runs under the RG with anomalous dimension $\eta$:

$$\frac{d\ln G}{d\ln k} = -\eta, \qquad G(k) = G_{\text{N}}\left(\frac{k_*}{k}\right)^\eta \quad \text{for } k \ll k_*, \tag{4}$$

where $k_*$ is the crossover momentum and $G_{\text{N}} = G(k_*)$ is the measured Newton constant.

The modified Newtonian potential in momentum space is

$$\tilde{\Phi}(k) = -\frac{4\pi G(k)\,M}{k^2} = -\frac{4\pi G_{\text{N}} M\,k_*^\eta}{k^{2+\eta}}. \tag{5}$$

Fourier-transforming to position space in $d = 3$ spatial dimensions:

$$\Phi(r) = -M\int \frac{d^3k}{(2\pi)^3}\,\frac{4\pi G(k)}{k^2}\,e^{i\boldsymbol{k}\cdot\boldsymbol{r}}. \tag{6}$$

### C. Why $\eta = 1$ is the marginal case

The integral (6) with $G(k) \propto k^{-\eta}$ yields three qualitatively distinct behaviors:

| $\eta$ | $\Phi(r)$ behavior | Physical status |
|---|---|---|
| $\eta < 1$ | $r^{\eta-1}$ (decaying) | IR-irrelevant |
| $\eta = 1$ | $\ln r$ (logarithmic) | **Marginal, scale-invariant** |
| $\eta > 1$ | $r^{\eta-1}$ (growing) | Violates solar-system constraints |

For $\eta = 1$, the running becomes $G(k) = G_{\rm N}(1 + k_*/k)$, and the explicit position-space results are:

$$\Phi(r) = -\frac{G_{\rm N} M}{r} + \frac{2G_{\rm N} M k_*}{\pi} \ln\left(\frac{r}{r_0}\right), \tag{7}$$

$$F(r) = -\frac{G_{\rm N} M}{r^2} - \frac{2G_{\rm N} M k_*}{\pi\, r}, \tag{8}$$

$$v^2(r) = r|F(r)| = \frac{G_{\rm N} M}{r} + \frac{2G_{\rm N} M k_*}{\pi}. \tag{9}$$

At $r \gg r_c = 1/k_*$, the second term dominates and

$$v^2(r) \approx V_0^2 = \frac{2G_{\rm N} M_{\rm bar}\, k_*}{\pi} = \text{const}, \tag{10}$$

yielding a *flat rotation curve*.

### D. Connection to $\Sigma_{\rm grav}$

Using $\Sigma_{\rm grav} = 2|\Phi|/c^2$ (Paper 2) with the running potential (7):

$$\Sigma_{\rm grav}(r) = \frac{r_{\rm s}}{r} + \frac{4k_* r_{\rm s}}{\pi} \ln\left(\frac{r}{r_0}\right), \tag{11}$$

where $r_{\rm s} = 2G_{\rm N} M/c^2$. This is *not* an ad hoc addition. It is the same $\Sigma = -\ln(-g_{00})$ from Paper 2, evaluated with the quantum-corrected potential that follows from the RG-improved gravitational channel.

### E. SPARC galaxy rotation curve data

Three SPARC galaxies fitted by Kumar [2] with $\eta = 1$ give a *universal* crossover scale:

| Galaxy | $M_{\rm bar}\,[M_\odot]$ | $V_0\,[\mathrm{km/s}]$ | $k_*\,[\mathrm{kpc}^{-1}]$ |
|---|---|---|---|
| S:700624 | $1.01\times10^{12}$ | $271.4\pm2.6$ | $(2.66\pm0.05)\times10^{-2}$ |
| S:700916 | $5.85\times10^{11}$ | $210.4\pm2.4$ | $(2.76\pm0.06)\times10^{-2}$ |
| S:705253 | $3.55\times10^{11}$ | $158.9\pm1.9$ | $(2.60\pm0.06)\times10^{-2}$ |

The values of $k_*$ are consistent across a factor $\sim 3$ in baryonic mass, giving $r_c \approx 37\,\mathrm{kpc}$. NFW dark matter halos have no analogous universal scale.

Gubitosi, Piattella, and Casarini [3] performed MCMC fits to 100 SPARC galaxies using the RGGR model ($G(\mu) = G_0/[1+\nu\ln(\mu^2/\mu_0^2)]$) and found that the running-$G$ framework is *competitive with NFW*: 47 galaxies prefer RGGR, 40 prefer NFW, and 13 are inconclusive.

## II. S2. OBSERVER-DEPENDENT $\Sigma_{\rm DM}$ CALCULATION

### A. Classical and quantum observer algebras

**Definition 1** (Observer algebras)**.** *For a gravitating system with total algebra $\mathcal{A}_{\rm total}$, define:*

- $\mathcal{A}_{\rm classical}$*: the subalgebra of electromagnetic observables (photon frequencies, luminosities, positions from EM detection).*
- $\mathcal{A}_{\rm quantum} = \mathcal{A}_{\rm total}$*: the full algebra including gravitational entanglement structure.*

By construction, $\mathcal{A}_{\rm classical} \subset \mathcal{A}_{\rm quantum}$. The restriction map $\mathcal{N}_{\rm cl} = E_{\mathcal{A}_{\rm classical}}$ is a quantum channel (conditional expectation).

### B. Computing the two entropy productions

By the DPI/monotonicity theorem (Paper 1, Theorem 1 of the observer-dependent framework):

$$\Sigma_{\rm classical} \geq \Sigma_{\rm quantum}. \tag{12}$$

For a point mass $M$ in the weak-field regime:

$$\Sigma_{\rm quantum}(r) = \frac{r_{\rm s}}{r}, \tag{13}$$

$$\Sigma_{\rm classical}(r) = \frac{r_{\rm s}}{r} + \frac{4k_* r_{\rm s}}{\pi}\ln\left(\frac{r}{r_0}\right) + \frac{r}{L_{\rm dS}}, \tag{14}$$

where the additional terms arise because the classical observer cannot access the gravitational entanglement that would reduce the apparent information loss.

### C. Dark matter as the observer gap

**Definition 2** (Dark matter entropy production)**.**

$$\Sigma_{\mathrm{DM}}(r) \equiv \Sigma_{\mathrm{classical}}(r) - \Sigma_{\mathrm{quantum}}(r) = \frac{4k_* r_{\mathrm{s}}}{\pi} \ln\left(\frac{r}{r_0}\right) + \frac{r}{L_{\mathrm{dS}}}. \tag{15}$$

The corresponding temporal asymmetry parameter for dark matter is

$$\tau_{\mathrm{DM}} = 1 - \exp(-\Sigma_{\mathrm{DM}}/2). \tag{16}$$

### D. Numerical estimate at the solar radius

For the Milky Way ($M \approx 6 \times 10^{10}\, M_\odot$) at $r = 8\,\mathrm{kpc}$:

$$\begin{aligned}
\Sigma_{\mathrm{Newton}} &= \frac{r_{\mathrm{s}}}{r} \approx 3 \times 10^{-6}, \\
\Sigma_{\mathrm{running}} &\approx \frac{4k_* r_{\mathrm{s}}}{\pi} \ln\left(\frac{8}{37}\right) \approx -1 \times 10^{-6}, \\
\Sigma_{\mathrm{DM}} &\approx 3.5 \times 10^{-7}, \\
\tau_{\mathrm{DM}} &\approx 1.7 \times 10^{-7}.
\end{aligned} \tag{17}$$

This tiny but nonzero $\tau_{\mathrm{DM}}$ is the origin of the galactic "dark matter" at this radius: it represents the information inaccessible to a classical (electromagnetic) observer.

## III. S3. CROOKS $\to a_0$ DERIVATION: EIGHT ROUTES

We systematically attempt to derive the MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s}^2$ from the Crooks fluctuation theorem [6] within the $\tau$ framework.

### A. Numerical benchmarks

With $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}} = 2.27 \times 10^{-18}\,\mathrm{s^{-1}}$:

$$\begin{aligned} cH_0 &= 6.80 \times 10^{-10}\,\mathrm{m/s^2}, \\ cH_0/(2\pi) &= 1.08 \times 10^{-10}\,\mathrm{m/s^2}, \\ cH_0/6 &= 1.13 \times 10^{-10}\,\mathrm{m/s^2}, \\ a_0\,(\mathrm{MOND}) &= 1.20 \pm 0.02 \times 10^{-10}\,\mathrm{m/s^2}. \end{aligned} \tag{18}$$

### B. Route 1: Naive $\Sigma = 1$ transition (FAILS)

Setting $\Sigma_{\mathrm{grav}} = r_{\mathrm{s}}/r = 1$ gives $a = c^4/(4GM)$, which is mass-dependent. For the Milky Way, $a \sim 10^2\,\mathrm{m/s^2}$, a factor $10^{12}$ above $a_0$. The naive transition identifies the event horizon, not the MOND scale.

### C. Route 2: $\Sigma_{\mathrm{grav}} = \Sigma_{\mathrm{cosmo}}$ (FAILS)

Balancing $r_{\mathrm{s}}/r = r^2/r_H^2$ gives $r_{\mathrm{tr}} = (r_{\mathrm{s}}\, r_H^2)^{1/3}$ and $a_{\mathrm{tr}} = (GM)^{1/3} H_0^{4/3}/2^{2/3}$. This is still mass-dependent and $\sim 10^{-14}\,\mathrm{m/s^2}$.

### D. Route 3: Verlinde in $\tau$ language (WORKS, imported)

Verlinde's emergent gravity [4] gives

$$g_D(r) = \sqrt{\frac{cH_0\, g_B(r)}{6}}, \tag{19}$$

with transition $g_D = g_B$ at $a_0 = cH_0/6 \approx 1.13 \times 10^{-10}\,\mathrm{m/s^2}$.

In $\tau$ language: in the Newtonian regime ($g \gg a_0$), $\Sigma \approx r_{\mathrm{s}}/r$ and $\tau \approx r_{\mathrm{s}}/(2r)$; in the MOND regime ($g \ll a_0$), $\Sigma_{\mathrm{eff}}$ grows as $\ln r$ from the displaced de Sitter entanglement, and $\tau$ exceeds the Newtonian prediction. However, the factor 6 is imported from Verlinde's specific geometric assumptions ($V/A = r/3$, elastic factor 1/2) rather than derived from Crooks.

### E. Route 4: Unruh–de Sitter matching (FAILS by $\sim 6\times$)

Setting $T_U = T_{\mathrm{dS}}$ gives $a = cH_0 \approx 6.8 \times 10^{-10}\,\mathrm{m/s^2}$, a factor $\sim 6$ too large.

### F. Route 5: Entropic force matching (circular)

Setting the Verlinde dark force equal to the Newtonian force reproduces $a_0 = cH_0/6$, but the calculation imports Verlinde's specific formula, making it circular.

### G. Route 6: Volume-law entropy with Crooks (INCOMPLETE)

The displaced Crooks entropy $\Sigma_{\text{displaced}} \sim S_V\, \epsilon^2/2$ involves $S_V \propto Mc^2/(\hbar H_0)$ and $\epsilon \sim r_{\text{s}}/r$. The product is mass-dependent. To extract a universal $a_0$, one needs Verlinde's prescription for counting degrees of freedom on holographic screens.

### H. Route 7: Geometric factor derivation (circular)

The factor $6 = 3 \times 2$ in Verlinde's $a_0 = cH_0/6$ comes from: 3 from $V/A = r/3$ (spherical geometry), and 2 from the quadratic QRE near equilibrium $D(\rho + \delta \| \rho) \approx \frac{1}{2}\text{Tr}(\delta^2/\rho)$. These factors are *geometrically motivated* from the $\tau$ framework but not derived from Crooks.

### I. Route 8: KMS–Crooks connection (BEST)

The de Sitter vacuum satisfies the KMS condition with thermal period $\beta_{\text{dS}} = 2\pi/H_0$. The spatial thermal wavelength is $\lambda_{\text{dS}} = c\beta_{\text{dS}} = 2\pi c/H_0$, and the corresponding acceleration is

$$a_0 = \frac{c^2}{\lambda_{\text{dS}}} = \frac{c^2}{2\pi c/H_0} = \frac{cH_0}{2\pi} \approx 1.08 \times 10^{-10}\,\text{m/s}^2, \tag{20}$$

within 10% of the empirical value.

The physical logic:

1. The KMS condition is exact for the de Sitter equilibrium state.

2. The Crooks theorem requires a thermal state; KMS is the mathematical definition of such a state in QFT.

3. The modular flow defines "thermal time," and Crooks entropy production $\Sigma$ is measured in these units.

4. The MOND transition $\Sigma \sim 1$ occurs when the gravitational acceleration matches the de Sitter thermal scale.

This reframes MOND as *marginal irreversibility*: the transition from $\Sigma \gg 1$ (Newtonian, strongly irreversible) to $\Sigma \ll 1$ (deep MOND, nearly reversible).

### J. Summary of routes

| Route | Method | Result | Verdict |
|---|---|---|---|
| 1 | Naive $\Sigma = 1$ | Mass-dep. | FAILS |
| 2 | $\Sigma_{\text{grav}} = \Sigma_{\text{cosmo}}$ | Mass-dep. | FAILS |
| 3 | Verlinde in $\tau$ | $cH_0/6$ | Works (imported) |
| 4 | Unruh–dS match | $cH_0$ | FAILS ($\times 6$) |
| 5 | Entropic force | $cH_0/6$ | Circular |
| 6 | Volume-law Crooks | Incomplete | — |
| 7 | Geometric factor | $cH_0/6$ | Circular |
| 8 | KMS–Crooks | $cH_0/(2\pi)$ | **Best** |

All routes agree: $a_0 \sim cH_0/\mathcal{O}(1)$, with the $\mathcal{O}(1)$ factor between 1 and $2\pi$.

## IV. S4. BULLET CLUSTER: ENTANGLEMENT RIGIDITY

### A. The challenge

The Bullet Cluster (1E 0657-558) shows an offset between the gravitational lensing center (tracing total mass) and the X-ray center (tracing hot gas). In ΛCDM, this arises because collisionless dark matter passes through while gas is ram-pressure stripped. Any modified gravity theory must explain this offset.

### B. Timescale estimates

In the $\tau$ framework, the "dark component" is gravitational entanglement of the stellar/galactic degrees of freedom. Define two timescales:

$$t_{\rm ent} = \frac{r_{\rm gal}}{c} \approx \frac{30\,{\rm kpc}}{c} \approx 10^5\,{\rm yr}, \tag{21}$$

$$t_{\rm coll} \approx \frac{D_{\rm sep}}{v_{\rm rel}} \approx \frac{700\,{\rm kpc}}{4700\,{\rm km/s}} \approx 1.5 \times 10^8\,{\rm yr}, \tag{22}$$

where $t_{\rm ent}$ is the light-crossing time for the galaxy's entanglement region and $t_{\rm coll}$ is the cluster crossing time.

Since $t_{\rm coll} \gg t_{\rm ent}$, the entanglement structure can respond (approximately) adiabatically to the collision dynamics. The entanglement entropy is anchored to the stellar distribution (which is collisionless), not to the gas (which is collisional).

### C. Why entanglement stays with galaxies

The running-$G$ contribution to $\Sigma$ depends on the matter distribution through $M_{\rm bar}(r)$. Stars and dark matter (in the conventional picture) are collisionless; gas is collisional. In the $\tau$ framework:

$$\Sigma_{\rm total}(r) = \underbrace{\frac{r_{\rm s}^{\rm stars}}{r} + \frac{4k_* r_{\rm s}^{\rm stars}}{\pi} \ln r}_{\text{follows stars}} + \underbrace{\frac{r_{\rm s}^{\rm gas}}{r}}_{\text{follows gas}} \quad . \tag{23}$$

The logarithmic (running-$G$) correction is dominated by the stellar mass distribution, which passes through the collision unimpeded. The gas contribution to the Newtonian term is subdominant ($M_{\rm gas} \ll M_{\rm stars}$ in the subcluster galaxies).

### D. Quantitative lensing-gas offset bound

The predicted lensing center is at the center of mass of $\Sigma_{\rm total}$. The gas displacement $\Delta x_{\rm gas} \approx 200$–$300\,{\rm kpc}$ (observed) produces a shift in the lensing center of order

$$\Delta x_{\rm lens} \approx \frac{M_{\rm gas}}{M_{\rm total}}\,\Delta x_{\rm gas} \sim 0.1\,\Delta x_{\rm gas} \sim 20\text{–}30\,{\rm kpc}, \tag{24}$$

consistent with the observed lensing-gas offset of $\sim 150\,{\rm kpc}$ only if the running-$G$ contribution provides the remaining "collisionless" component. This requires $\Sigma_{\rm running} \gtrsim 4\,\Sigma_{\rm Newton}$ at the cluster scale, which remains to be quantitatively verified.

## V. S5. CROOKS INTERPOLATION FUNCTION

### A. Derivation from $P(\Sigma)/P(-\Sigma) = e^{\Sigma}$

The Crooks relation gives $P_F(\Sigma)/P_R(-\Sigma) = e^{\Sigma}$. If we identify $\Sigma = a/a_0$ (the local acceleration in units of the MOND scale), then the ratio of forward to reverse probabilities is $\exp(a/a_0)$, and the "efficiency" of the gravitational process is

$$\mu_{\text{Crooks}}(x) = 1 - e^{-x}, \qquad x = \frac{a}{a_0} = \frac{g_{\text{bar}}}{a_0}. \tag{25}$$

Properties:

- $x \gg 1$: $\mu \approx 1$ (Newtonian regime).
- $x \ll 1$: $\mu \approx x$ (deep MOND, linear).
- The transition is at $x \sim 1$ ($a \sim a_0$).

### B. Comparison with standard MOND interpolation functions

| Function | $\mu(x)$ | $x = 0.1$ | $x = 1$ | $x = 3$ |
|---|---|---|---|---|
| Standard | $x/(1+x)$ | 0.091 | 0.500 | 0.750 |
| Simple | $x/\sqrt{1+x^2}$ | 0.100 | 0.707 | 0.949 |
| RAR (empirical) | $1 - e^{-\sqrt{x}}$ | 0.272 | 0.632 | 0.826 |
| Crooks-inspired | $1 - e^{-x}$ | 0.095 | 0.632 | 0.950 |

The Crooks-inspired form transitions faster than the "standard" function but matches the "simple" function well for $x \geq 1$. Its behavior at $x = 1$ ($\mu = 1 - 1/e \approx 0.632$) coincides with the empirical RAR interpolation $1 - e^{-\sqrt{x}}$ evaluated at $x = 1$.

### C. Rotation curve predictions

The effective gravitational acceleration is

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{\mu(g_{\text{bar}}/a_0)} = \frac{g_{\text{bar}}}{1 - \exp(-g_{\text{bar}}/a_0)}. \tag{26}$$

For a galaxy with baryonic mass profile $M_{\rm bar}(r)$, the circular velocity is $v^2(r) = r\, g_{\rm obs}(r)$, reproducing flat rotation curves in the regime $g_{\rm bar} \ll a_0$. Direct comparison with SPARC data requires fitting $\Upsilon_*$ (stellar mass-to-light ratio) as the sole free parameter.

## VI. S6. DARK ENERGY FROM COSMIC INFORMATION (COSMIN)

### A. Padmanabhan's argument

Padmanabhan [7] argues that the cosmological constant is determined by the requirement that the total cosmic information be finite:

$$I_{\rm CosMIn} = \int_0^\infty (N_{\rm sur}(t) - N_{\rm bulk}(t))\, dt < \infty, \tag{27}$$

where $N_{\rm sur} = A/(l_P^2)$ and $N_{\rm bulk}$ are the surface and bulk degrees of freedom of the Hubble sphere.

### B. Translation to the $\tau$ framework

In our language:

$$I_{\rm CosMIn} = \int_0^\infty \tau(t)\, dt < \infty, \tag{28}$$

where $\tau(t)$ is the cosmological temporal asymmetry at time $t$. The requirement that this integral converges demands $\tau(t) \to 0$ sufficiently fast as $t \to \infty$, which requires accelerated expansion ($\Lambda > 0$).

### C. Connection to the cosmological horizon

At the de Sitter future ($\Lambda$-dominated era), the Hubble sphere approaches $r_H = c/H_\Lambda$ with $H_\Lambda = \sqrt{\Lambda/3}$. The horizon entropy is $S_H = \pi r_H^2/(G_{\rm N}\hbar)$, and the Kodama-vector $\tau$ profile is

$$\tau(r) = 1 - \sqrt{1 - H_\Lambda^2 r^2/c^2}, \quad r < c/H_\Lambda. \tag{29}$$

This goes from $\tau = 0$ at the observer to $\tau = 1$ at the horizon, confirming that the cosmological horizon is the maximum-$\tau$ boundary.

# Part II: Paper 4 — Grand Unification via $\Sigma = D(\rho_{\rm st}\|\rho_{\rm m})$

## VII. S7. BACKGROUND QRE ON FRW

### A. Cai–Kim apparent horizon thermodynamics

For flat FRW with scale factor $a(t)$ and Hubble parameter $H = \dot{a}/a$, the apparent horizon is at $r_A = 1/H$. Its thermodynamic quantities are [8]:

$$T_A = \frac{H}{2\pi}, \quad S_A = \frac{\pi}{G_{\rm N} H^2}, \quad E = \frac{1}{2G_{\rm N} H}. \tag{30}$$

### B. $D_{\rm background} = 0$ implies the Friedmann equations

Define the spacetime thermal state and matter state by

$$\rho_{\rm st} = \frac{1}{Z_A} \exp\Big( - \frac{2\pi}{H}\, M(r_A)\Big), \tag{31}$$

$$\rho_{\rm m} = \text{state encoding } T_{ab} \text{ within } r_A. \tag{32}$$

The QRE between them: $D(\rho_{\rm st}\|\rho_{\rm m}) = \beta_A\, \Delta E - \Delta S$. At equilibrium (Friedmann equations satisfied), $D = 0$: the spacetime and matter states are in "entanglement equilibrium" [9].

### C. Detailed derivation

Applying the Clausius relation $-dE = T_A\, dS_A$ at the apparent horizon:

*Energy*: $E = \rho \cdot (4\pi/3) r_A^3 = 4\pi\rho/(3H^3)$.

*Differentiating*: Using $\dot{\rho} = -3H(\rho + p)$:

$$\dot{E} = \frac{4\pi}{3}\left[\frac{\dot{\rho}}{H^3} - \frac{3\rho\dot{H}}{H^4}\right] = \frac{4\pi}{3}\left[\frac{-3H(\rho+p)}{H^3} - \frac{3\rho\dot{H}}{H^4}\right]. \tag{33}$$

*Entropy*: $S_A = \pi/(G_{\rm N} H^2)$, so $\dot{S}_A = -2\pi\dot{H}/(G_{\rm N} H^3)$.

*Clausius relation*: $-\dot{E} = T_A \dot{S}_A$ gives

$$4\pi \left[ \frac{\rho + p}{H^2} + \frac{\rho \dot{H}}{H^3} \right] = \frac{\dot{H}}{G_{\mathrm{N}} H^2}. \tag{34}$$

This yields:

$$\dot{H} = -4\pi G_{\mathrm{N}}(\rho + p) \quad \text{(second Friedmann equation)}, \tag{35}$$

$$H^2 = \frac{8\pi G_{\mathrm{N}}}{3}\rho \quad \text{(first Friedmann equation, from the constraint)}. \tag{36}$$

### D. Consistency check: $\Sigma_{\mathrm{grav}} = 0$ on FRW

In comoving coordinates, $g_{00} = -1$, so $\Sigma_{\mathrm{grav}} = -\ln(-g_{00}) = 0$. This is consistent: a comoving observer experiences zero gravitational entropy production. The cosmological arrow of time arises entirely from expansion (increasing scale factor), not from gravitational information loss.

## VIII. S8. FIRST-ORDER QRE PERTURBATION

### A. The Aalsma–Bak modular Hamiltonian perturbation

Following Aalsma and Bak [10], the modular Hamiltonian for the FRW causal diamond, perturbed by a Newtonian potential $\Phi$, becomes

$$\delta K = 2\, S_A\, \Phi, \tag{37}$$

where $S_A = \pi/(G_{\mathrm{N}} H^2)$ is the de Sitter (apparent horizon) entropy.

### B. Entanglement first law: $\delta D^{(1)} = 0$

The first-order variation of the QRE is given by the entanglement first law [11, 12]:

$$\delta D^{(1)} = \delta\langle K_A \rangle - \delta S_A. \tag{38}$$

The modular Hamiltonian perturbation gives $\delta\langle K_A \rangle = 2 S_A\, \Phi$.

The entropy perturbation is $\delta S_A = -S_A\, \delta_m$, where $\delta_m = \delta\rho/\rho$ is the matter density contrast (from the shift in the apparent horizon location).

Setting $\delta D^{(1)} = 0$ (entanglement equilibrium at first order):

$$2\Phi + \delta_m = 0 \quad \Longrightarrow \quad \Phi = -\frac{1}{2}\,\delta_m = -\frac{\delta\rho}{2\rho}. \tag{39}$$

This is the **Poisson equation in the sub-Hubble limit**: $k^2\Phi = -4\pi G_{\rm N} a^2 \rho\, \delta_m$, which for modes well inside the horizon ($k \gg H$) reduces to $\Phi \approx -\delta_m/2$.

### C. Gauge choices

The above is derived in Newtonian gauge: $ds^2 = -(1+2\Phi)\,dt^2 + a^2(1-2\Psi)\,\delta_{ij}\,dx^i\,dx^j$. For GR without anisotropic stress, $\Phi = \Psi$. The Bardeen potentials are gauge-invariant and coincide with $\Phi, \Psi$ in Newtonian gauge. In synchronous gauge, the calculation requires expressing $\delta K$ and $\delta S_A$ in terms of the synchronous potentials $h$ and $\eta$.

### D. What appears and what does not

At first order in the standard QRE formalism:

- **Present**: scalar metric perturbations $(\Phi, \Psi)$ and matter perturbations $(\delta\rho, \delta p, \delta q)$.
- **Absent**: any independent "observer field" or Khronon-like scalar. The Kodama vector, which plays the role of the observer, is determined by the metric—not an independent degree of freedom.

## IX. S9. MODULAR KHRONON EMERGENCE

### A. The modular time perturbation

The modular Hamiltonian $K$ generates a flow with modular parameter $s$. The "modular time" associated with metric perturbation $\Phi$ is

$$\delta T_{\rm mod} = \frac{\delta K}{2\pi H^2/(2\pi)} = \frac{2S_A\,\Phi}{H} = -\frac{2\Phi}{H}. \tag{40}$$

This perturbation defines a "displaced foliation"—the constant-modular-time surfaces are shifted from the constant-cosmic-time surfaces by $\delta T_{\rm mod}$.

### B. Dispersion relation

On a Minkowski background, the modular time perturbation $\delta T_{\rm mod}$ satisfies the constraint $\delta D^{(1)} = 0$, which ties $\delta T_{\rm mod}$ to $\Phi$. The perturbation does *not* satisfy a wave equation; instead, it is determined algebraically by the Poisson equation (39). Its effective dispersion relation is

$$\omega = 0 \quad \text{(non-propagating)}, \tag{41}$$

because the modular-time perturbation is a *constraint mode*, not a propagating degree of freedom.

In a matter-dominated universe, the Newtonian potential is constant ($\dot{\Phi} = 0$), so $\delta T_{\rm mod}$ is frozen—it grows in place without oscillating, exactly like pressureless dust (CDM).

### C. Why $c_s \neq 0$ in the radiation era

During radiation domination, the gravitational potential decays ($\dot{\Phi} \neq 0$), so $\delta T_{\rm mod}$ is driven. The effective sound speed $c_s^2 = \delta p/\delta\rho$ for the modular perturbation is

$$c_s^2 \approx \begin{cases} 0 & \text{matter era } (w = 0), \\ \mathcal{O}(k/\mu)^2 & \text{radiation era, subleading.} \end{cases} \tag{42}$$

For all CMB-relevant scales ($k \ll \mu$ if $\mu^{-1} \sim$ kpc), $c_s^2 \approx 0$ to excellent approximation.

### D. Effective mass

The Khronon mass parameter $\mu$ sets the scale at which the non-propagating behavior breaks down. From the Blanchet–Skordis theory [13]:

$$\mu_{\rm eff} \sim 2\pi H_0 \approx 1.4 \times 10^{-17}\,\mathrm{s}^{-1}. \tag{43}$$

In position space, $\mu^{-1} \sim 223\,$kpc, consistent with the crossover between MOND and cosmological regimes.

## X. S10. OBSERVER ENTANGLEMENT EQUILIBRIUM (OEE)

### A. Formal statement

**Definition 3** (Observer Entanglement Equilibrium)**.** *The physical state $\omega$ satisfies OEE if the QRE is stationary with respect to variations in the metric* and *the observer's 4-velocity:*

$$\frac{\delta D}{\delta g_{ab}} = 0, \qquad \frac{\delta D}{\delta u^a} = 0, \tag{44}$$

*subject to $g_{ab}u^a u^b = -1$ (enforced by a Lagrange multiplier $\lambda$).*

The first equation gives the Einstein equations (Dorau-Much/Jacobson). The second is new: it determines the observer's 4-velocity as a *dynamical* field.

### B. Derivation of the Khronon field equation from OEE

The QRE depends on $u^a$ through the modular Hamiltonian $K[u] = (2\pi/\kappa_u) \int T_{ab}\, u^a\, d\Sigma^b$. Varying with respect to $u^a$ (with the constraint $u^a u_a = -1$) yields

$$(g_{ab} + u_a u_b)\, \nabla^b \left( \frac{\partial K}{\partial u^a} \right) = 0, \tag{45}$$

which, for $u^a = -(c/Q)\nabla^a T$ (the hypersurface-orthogonal form), reduces to the Khronon field equation with kinetic function

$$K(Q) = \mu^2 (Q-1)^2. \tag{46}$$

### C. Connection to Petz optimality

The Petz optimality argument: the kinetic function $K(Q)$ must have a quadratic minimum at the background value $Q_0 = 1$, because:

1. Petz optimality requires the channel to be maximally recoverable.
2. Maximal recoverability implies minimal entropy production for perturbations.
3. Minimal entropy production requires perturbations not to propagate ($c_s = 0$).
4. $c_s = 0$ requires $K(Q)$ to have a quadratic minimum at the background: $K(Q) \sim \mu^2 (Q - Q_0)^2$.

This selects exactly the Blanchet–Skordis kinetic term [13].

### D. Why OEE gives $K(Q) = \mu^2(Q-1)^2$

The variation $\delta D/\delta u^a = 0$ at the background ($u^a$ = comoving) gives $K'(Q_0) = 0$. Combined with stability ($K''(Q_0) > 0$), the leading-order kinetic function is

$$K(Q) = K(Q_0) + \tfrac{1}{2}K''(Q_0)(Q-Q_0)^2 + \cdots = \mu^2(Q-1)^2 + \mathcal{O}((Q-1)^3). \tag{47}$$

## XI. S11. $\tau \to$ AEST MAPPING

### A. Complete AeST action

The Skordis–Zlosnik AeST action [14] is

$$S = \int d^4x \, \frac{\sqrt{-g}}{16\pi\tilde{G}} \Big[ R - \frac{K_B}{2} F_{\mu\nu}F^{\mu\nu} + 2(2-K_B)J^\mu \nabla_\mu \phi - (2-K_B)Y - \mathcal{F}(Y,Q) - \lambda(A^\mu A_\mu + 1) \Big] + S_{\rm m}, \tag{48}$$

where $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$, $J^\mu = A^\nu \nabla_\nu A^\mu$ (aether acceleration), $Y = D^\mu\phi \, D_\mu\phi$ (spatial scalar kinetic term with $D_\mu\phi = (\delta^\nu_\mu + A^\nu A_\mu)\nabla_\nu\phi$), and $Q = A^\mu \nabla_\mu \phi$ (temporal scalar derivative).

### B. The identification: $u^a = A^\mu$

The structural identification is exact:

| $\tau$ framework | AeST / Khronon | Math. object |
|---|---|---|
| Observer $u^a$ | Aether $A^\mu$ | Unit timelike vector |
| Time function $T$ | Khronon $\tau$ | Scalar foliation |
| $u^a u_a = -1$ | $A^\mu A_\mu = -1$ | Same constraint |
| $\Sigma = -\ln(-g_{00})$ | $\phi_s$ | Gravitational scalar |
| DPI monotonicity | MOND interpolation | Running of channel |

### C. What the $\tau$ framework determines vs. what is free

**Determined by $\tau$:**

1. The existence of a dynamical unit timelike field $u^a$ (from observer-dependence of $\tau$).

2. The quadratic-minimum form of the kinetic function $K(Q)$ (from Petz optimality; Section S10).

3. The sign constraints $c_a = c_1 + c_4 \geq 0$ (from DPI/energy conditions).

4. The constraint $c_1 + c_3 = 0$ (from $c_T = c$; GW170817 [16]).

**Free parameters (not determined)**:

1. The coupling constants $c_1$ and $c_4$ (two free parameters).

2. The mass scale $\mu$ (sets the DM energy density).

3. The MOND acceleration scale $a_0$ (entered via $\mathcal{F}(Y, Q)$).

4. The DBI completion (behavior away from the quadratic minimum).

### D. The stealth black hole tension

AeST admits "stealth" black hole solutions with exact Schwarzschild geometry [15], while the $\tau$ framework predicts the exponential metric $g_{00} = -e^{-r_s/r}$ (Paper 2). These are distinct and observationally distinguishable. This tension is real and must be acknowledged; its resolution requires determining whether the Petz saturation ansatz or the AeST field equations take precedence in the strong-field regime.

## XII. S12. FIVE LIMITS OF THE MASTER EQUATION

The master equation $\Sigma = D(\rho_{\rm st} \| \rho_{\rm m})$ reduces to known physics under different boundary conditions.

### A. Paper 1 limit: Quantum information

*Setting*: Flat spacetime ($g_{\mu\nu} = \eta_{\mu\nu}$), CPTP map $\mathcal{N}$, finite-dimensional Hilbert space. $\Sigma_{\rm grav} = 0$ (no gravitational contribution). The remaining $\Sigma$ is

$$\Sigma_{\rm QI} = D(\rho \| \sigma) - D(\mathcal{N}(\rho) \| \mathcal{N}(\sigma)). \tag{49}$$

The Petz recovery bound $F \geq \exp(-\Sigma_{\mathrm{QI}}/2)$, the equivalence chain $\tau = 0 \Leftrightarrow \Sigma = 0 \Leftrightarrow$ quantum Markov $\Leftrightarrow$ Knill–Laflamme, and the retrodiction Landauer principle all follow.

*Precision*: EXACT. No approximations.

### B. Paper 2 limit: Strong-field gravity

*Setting*: Static, spherically symmetric, asymptotically flat, $r \sim r_{\mathrm{s}}$.

The gravitational redshift channel gives $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$. For the exponential metric (exact Petz saturation): $g_{00} = -\exp(-r_{\mathrm{s}}/r)$, hence $\Sigma = r_{\mathrm{s}}/r$. Three independent first-principles routes confirm this (modular flow, gravitational Landauer, pure-loss bosonic channel).

*Precision*: Steps 1–3 DERIVED; exact saturation CONJECTURED.

### C. Paper 3 limit: Galactic dynamics

*Setting*: Weak field, $r \sim$ kpc, RG-improved channel.

The DPI applied to the RG flow (Casini–Huerta [1]) gives monotonicity $\Sigma(k_{\mathrm{IR}}) \geq \Sigma(k_{\mathrm{UV}})$. With $\eta = 1$ running (Kumar [2]):

$$\Sigma_{\mathrm{weak}}(r) = \frac{r_{\mathrm{s}}}{r} + \frac{4k_* r_{\mathrm{s}}}{\pi} \ln\left(\frac{r}{r_0}\right). \tag{50}$$

This produces flat rotation curves, the radial acceleration relation, and the baryonic Tully-Fisher relation.

*Precision*: Running $G$ KNOWN (asymptotic safety); $\eta = 1$ PHENOMENOLOGICAL (not derived from $\tau$ framework).

### D. Paper 5 limit: Observer-dependent time

*Setting*: General system, observer $O$ with algebra $\mathcal{A}_O \subset \mathcal{A}_{\mathrm{total}}$.

The observer's effective channel is $\mathcal{N}_O : \mathcal{A}_{\mathrm{total}} \to \mathcal{A}_O$ (conditional expectation). Observer-dependent entropy production:

$$\Sigma_O = D(\omega \| \omega_{\mathrm{ref}}) - D(\mathcal{N}_O(\omega) \| \mathcal{N}_O(\omega_{\mathrm{ref}})), \tag{51}$$

with monotonicity: $O_1 \leq O_2 \Rightarrow \Sigma_{O_1} \geq \Sigma_{O_2}$.

*Precision*: Monotonicity PROVED (DPI).

### E. Friedmann limit: Cosmological expansion

*Setting*: Homogeneous, isotropic FRW, $r \sim c/H_0$.

At the background level, $D = 0$ yields the Friedmann equations (Section S7). At first order, $\delta D^{(1)} = 0$ yields the Poisson equation (Section S8). The Kodama-vector $\tau$ profile $\tau(r) = 1 - \sqrt{1 - H^2 r^2/c^2}$ interpolates from $\tau = 0$ at the observer to $\tau = 1$ at the apparent horizon.

*Precision*: Background PROVED (Cai-Kim [8]); perturbations NEW SYNTHESIS (Aalsma–Bak [10]).

## XIII. S13. WHY $\Omega_{\mathrm{DM}}$ CANNOT BE DERIVED

### A. The fundamental obstruction

The JRSWW bound $F \geq \exp(-\Sigma/2)$ is a *universal* information-theoretic statement. It holds for every quantum channel, every pair of states, and every Hilbert space dimension. It knows nothing about:

- the specific matter content of the universe,
- the baryon-to-photon ratio,
- Big Bang nucleosynthesis,
- freeze-out temperatures.

Meanwhile, $\Omega_{\mathrm{DM}} h^2 \approx 0.12$ is a *contingent* fact about our universe. **A universal bound cannot determine a contingent parameter.**

### B. Summary of 10 failed attempts

| # | Approach | Verdict |
|---|---|---|
| 1 | $\Sigma = -\ln(g_{00})$ at fixed $r/r_H$ | DEAD END (arbitrary $r$) |
| 2 | $\tau = \Omega_{\rm DM}$ | DEAD END (tautological) |
| 3 | Numerology | DEAD END |
| 4 | Padmanabhan $N_{\rm sur}$ vs $N_{\rm bulk}$ | Predicts $\Lambda$, not $\Omega_{\rm DM}$ |
| 5 | Cumulative $\Sigma$ at $r_H$ | CIRCULAR ($\Sigma \sim \Omega_m$) |
| 6 | Bekenstein bound | DEAD END ($S/S_{\rm holo} \sim 10^{-18}$) |
| 7 | $\Sigma = 1$ at horizon | Suggestive ($\tau = 0.39$ vs $\Omega_m = 0.31$) |
| 8 | Holographic DM (Trivedi-Scherrer) | Indirect connection only |
| 9 | $N_{\rm bulk}/N_{\rm sur}$ ratio | CIRCULAR |
| 10 | Baryogenesis entropy | DEAD END |

### C. The correct analogy

$\Omega_{\rm DM}$ is to the $\tau$ framework as the mass of the Earth is to $F = ma$. Newton's second law provides the *framework* (how things respond to forces) but not the *initial conditions* (the mass of Earth). Similarly, $\Sigma = D(\rho_{\rm st}\|\rho_{\rm m})$ provides the *structure* of gravity and dark matter but not the *amplitude* $\Omega_{\rm DM}h^2 = 0.12$, which requires cosmological initial conditions.

## XIV. S14. COMPARISON WITH OTHER UNIFICATION PROGRAMS

### A. String theory / M-theory

| Feature | String theory | $\tau$ framework |
|---|---|---|
| Starting point | Strings/branes | QRE ($\Sigma$) |
| UV complete? | Yes | No |
| Background independent? | No (perturbative) | Yes (algebraic) |
| Predictions | Extra dims, SUSY | GW echoes, flat curves |
| Testable (2026)? | No | Yes |

The $\tau$ framework is an *organizational principle*, not a competitor to microscopic theories.

If string theory provides the UV completion, $\tau$ may provide the macroscopic organizational structure—analogous to thermodynamics vis-à-vis statistical mechanics.

### B. Loop quantum gravity

LQG quantizes geometry itself (spin networks, discrete area spectrum). The $\tau$ framework does not quantize geometry but uses information loss as the fundamental quantity. A potential connection: the discrete area spectrum in LQG would quantize $\Sigma_{\text{grav}}$ in Planck units, providing a UV cutoff.

### C. Entropic gravity (Verlinde)

Verlinde [4] derives gravity from holographic entropy (area + volume law). The $\tau$ framework shares the philosophy (gravity = entropy production) but uses QRE rather than force-based entropy arguments. Verlinde's volume-law contribution may be the cosmological-scale limit of our $\Sigma$, while our $\Sigma_{\text{grav}} = r_{\text{s}}/r$ is the strong-field limit. Ghari and Haghi [5] find Verlinde favored over MOND at $5.2\sigma$ for dwarf spheroidals.

### D. ER = EPR

The $\tau$ interpretation: a wormhole connecting two regions is a system for which the total observer (accessing both ends) has $\tau = 0$ (closed system, no arrow of time). An observer at one end only has $\tau > 0$ (the other end appears thermalized). This is the ER = EPR picture translated to $\tau$ language. Kaplan, Mittal, Penington, and Susskind (2024) proved this operationally for typical states.

### E. It from Qubit / Holographic QEC

The Petz recovery map appears in both frameworks:

- In holographic QEC, the Petz map reconstructs bulk operators from boundary data (Cotler et al., PRX 2019).
- In the $\tau$ framework, the Petz map is the optimal retrodiction (Paper 1).

These are the *same* mathematical object in different contexts. The $\tau$ framework extends beyond AdS/CFT (which requires a conformal boundary) because QRE is well-defined in any algebraic QFT.

### F. Summary: what type of theory $\tau$ is

The $\tau$ framework is not a microscopic theory (like string or LQG). It is a *structural theory*—universal predictions that hold for any microscopic completion satisfying the DPI. Its value:

1. Explains *why* Petz/QRE structures appear across domains.

2. Provides universal, model-independent predictions.

3. Connects QI, GR, cosmology, and philosophy of time.

4. Gives a computable definition of "arrow of time" at all scales.

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